SUPPORTING ADULT GUIDANCE

Combining the theorems

Circle Theorems · Lesson 9. For any supporting adult — teacher, practitioner, parent or carer.
You do not need to be a mathematics specialist to support this lesson.

What this lesson does

This lesson is synthesis: learners combine every circle theorem met so far in multi-step angle chases, and — crucially — justify each step by naming the theorem used. It introduces no new theorem; the demand is strategic. Learners meet a guided chase (each step validated for both value and reason), practise recognising which theorem a situation calls for, and watch a fully worked four-theorem chase. The habit being built is reasoned, defensible problem-solving.

Driving Question: *How do you crack a problem that needs several circle theorems at once?*

The mathematics, in plain terms

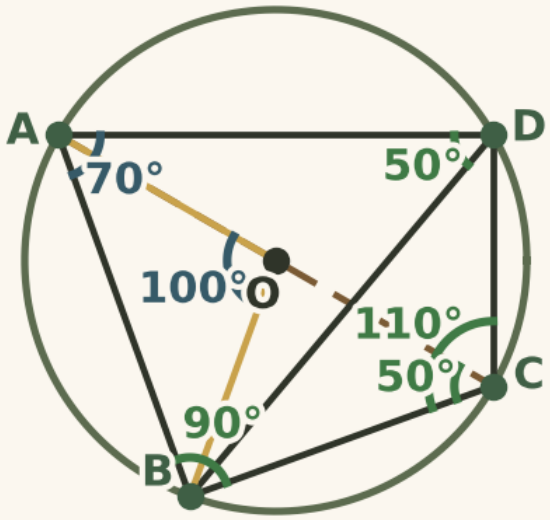
An angle chase proceeds from the givens by repeatedly applying whichever circle theorem is available, recording the reason at each step. In the worked figure (AC a diameter, $\angle AOB = 100°$): $\angle ACB = 50°$ (angle at centre is twice the angle at the circumference), $\angle ABC = 90°$ (angle in a semicircle), $\angle ADB = 50°$ (same segment), and with $\angle BAD = 70°$, $\angle BCD = 110°$ (cyclic quadrilateral, opposite angles supplementary). The order is flexible; naming the theorem is not optional.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — A three-step chase

In the circle, O is the centre, AC is a diameter, and the central angle ∠AOB = 100°. Find ∠ACB, ∠ABC and ∠ADB, giving a reason for each.

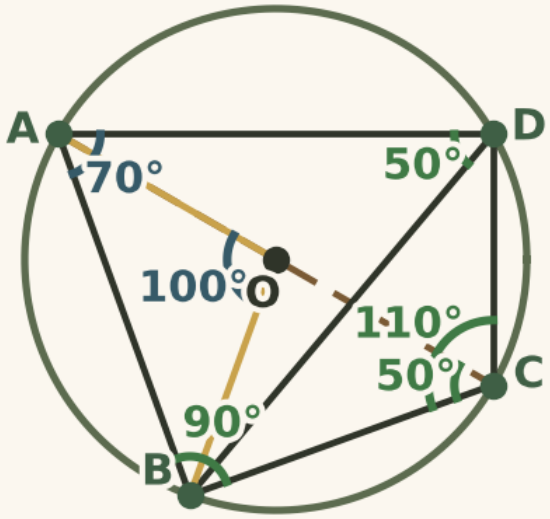


1. ∠ACB and ∠AOB stand on the same arc AB; the angle at the centre is twice the angle at the circumference.	∠ACB = 100 ÷ 2 = 50°
2. AC passes through O, so AC is a diameter; the angle in a semicircle is 90°.	∠ABC = 90°
3. ∠ADB and ∠ACB stand on the same chord AB from the same segment.	∠ADB = ∠ACB = 50°

∠ACB = 50°, ∠ABC = 90°, ∠ADB = 50° — each justified by a named theorem.

Example 2 — Extending the chase with a cyclic quadrilateral

In the same figure, ABCD is a cyclic quadrilateral and ∠BAD = 70°. Find the opposite angle ∠BCD.



1. ABCD is a cyclic quadrilateral, so its opposite angles add to 180°.	∠BCD + ∠BAD = 180°
2. Substitute the known angle ∠BAD = 70°.	∠BCD = 180 – 70

∠BCD = 110° — opposite angles of a cyclic quadrilateral are supplementary.

Value and reason, at every step

Insist that each step carries a named theorem — a correct value with no reason is not yet a solution, and the guided chase deliberately checks both. Encourage learners to plan a first move (the easiest reachable angle) rather than guessing at the target. Praise the moment one answer becomes the tool for the next; that is the essence of a chase. Keep to angles the theorems justify, and let learners build their own combined problems once secure. This consolidates the unit ahead of Lesson 10, which turns to formal proof and problem-solving.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	See how a chase starts from a reachable angle.	~8 min

Movement	Recognise which theorem each situation calls for.	~11 min
Reflection	Run a full guided chase, justifying every step.	~13 min
Creativity	Watch a four-theorem chase unfold.	~9 min
Rest	A short pause on holding the whole toolkit in mind.	~2 min
Nutrition	Intellectual: strategic, defensible reasoning.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Giving values with no reasons.
'A value with no reason is not a solution here — name the theorem for each step.'

Not knowing where to start.
'Find the easiest angle you can reach from the givens — a central angle to halve, or a diameter — and start there.'

Using the wrong theorem for a correct-looking value.
'Right number, but is that the theorem that justifies it? Check which angles actually stand on which arc.'

Stopping after one step.
'Good — now, what does that new angle let you find next?'

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson's Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: all the circle theorems so far: angle at the centre, angle in a semicircle, angles in the same segment, cyclic quadrilaterals, tangents, and the alternate segment theorem. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner freezes at the start, or gives answers with no reasons. The aim is a question that gets a first move and a named theorem.

- "Which angle can you find immediately from the givens? Start there."
- "What theorem justifies that step? Write it down."

- “Now you know that angle — what does it let you find next?”
- “Is that the theorem that justifies your value, or just a number that fits?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Where do I start an angle chase?
With the easiest angle you can reach from the givens — often halving a central angle, spotting a diameter (right angle), or matching two angles on the same chord. One reached angle usually unlocks the next.

Do I really have to write a reason every time?
Yes. In this unit a value with no reason is not a solution. The chain of named theorems is the argument; it is also what earns full marks in an exam.

What if two different theorems both seem to apply?
Either may be fine as long as it genuinely justifies that step. Choose the one whose conditions the diagram actually meets, and name it.

How do I get faster at choosing the right theorem?
Practise recognising the trigger for each: a central angle, a diameter, two angles on one chord, a cyclic quadrilateral, a tangent and chord. The Movement activity drills exactly this recognition.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.

Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.
Proficiency	Works through independently and explains why each step is true.
Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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